

Practice Questions

Test your comprehension by answering the following questions:

1. Do sets have a specific order for their elements? Do permutations? Do combinations?
2. How many permutations (without repetition) are there of a set of n elements?
3. How many combinations (without repetition) are there of a set of n elements?
4. Is $\{A, B, C\}$ a subset of $\{A, B, C\}$?
5. What is the formula for calculating n choose k , the number of possible combinations of k elements selected from a set of n elements?
6. Identify which of the following are permutations or combinations, with or without repetition:
 - a. AAA, AAB, AAC, ABA, ABB, ABC, ACA, ACB, ACC, BAA, BAB, BAC, BBA, BBB, BBC, BCA, BCB, BCC, CAA, CAB, CAC, CBA, CBB, CBC, CCA, CCB, CCC
 - b. ABC, ACB, BAC, BCA, CAB
 - c. (None), A, B, C, AB, AC, BC, ABC
 - d. (None), A, B, C, AA, AB, AC, BB, BC, CC, AAA, AAB, AAC, ABB, ABC, ACC, BBB, BBC, BCC, CCC
7. Draw a tree graph that can be used to generate all possible combinations of the set $\{A, B, C, D\}$.
8. Answer the three questions about recursive solutions for each of the recursive algorithms presented in this chapter:
 - a. What is the base case?
 - b. What argument is passed to the recursive function call?
 - c. How does this argument become closer to the base case?